

Unless otherwise stated, each problem is worth 4 points. Show all of your work.

Evaluate the following integrals.

1. [6 pts] $\int \ln 3x \, dx$

2. $\int \sin^2 3x \, dx$

3. $\int \cos^3 x \, dx$

4. $\int \sin^2 x \cos^2 x \, dx$

5. $\int \cos^3 x \sin x \, dx$

6. [6 pts] $\int \frac{dx}{\sqrt{4-x^2}}$

7. [8 pts] $\int_0^4 \sqrt{16-x^2} \, dx$

8. [2 pts] $\int \sqrt{16-x^2} \, dx$; hint: use the reference triangle and the double-angle identity, $\sin 2\theta = 2 \sin \theta \cos \theta$.